Electric Machines

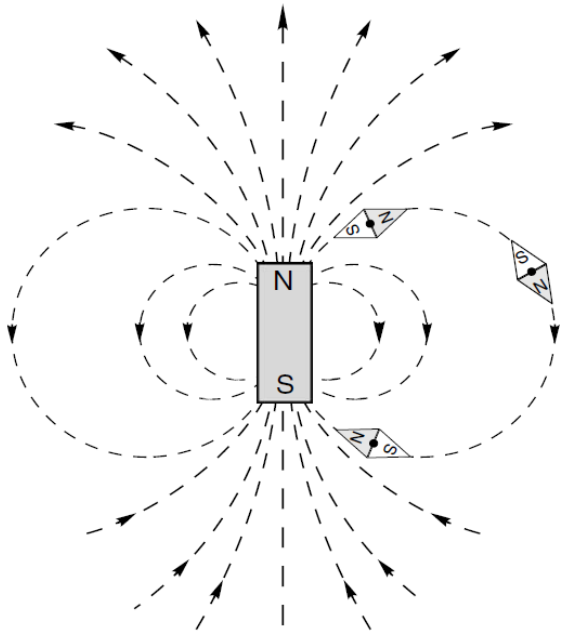
Magnetism Fundamentals – DC Motors

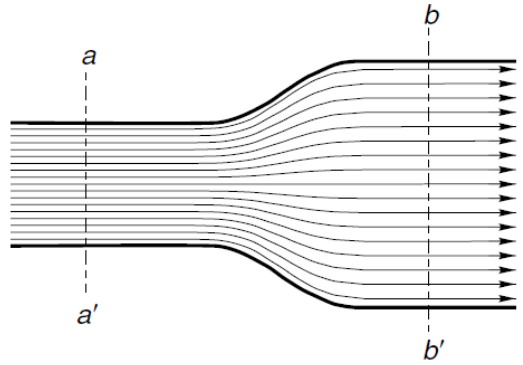
Objectives

- The student should
  - Describe the principles of electromagnetism: flux and flux density, Force on a Conductor, Torque on a coil (loop), magnetic dipole moment, magnetic fields created by currents, Faraday's law
  - Explain and describe the concept of reluctance, magnetomotive force, and magnetic circuits and how they are applied in motors.
  - Describe the DC and AC motor principles
  - Describe and explain the operating principles of DC motors: commutators and brushes.
  - Distinguish lap and wave windings.
  - Know and apply the conversion equations of a DC motor
  - Describe the classification of DC motors.

Magnetic Field and Magnetic Flux

- Magnetic field is an abstraction of an observable physical phenomenon.
- Magnetic flux quantifies the strength of the magnetic field. Pictorially, it's represented by lines with directions. The closer the lines, the stronger the magnetic flux.
- The unit of the magnetic flux (Φ) is the weber (Wb)
- The flux density (B) is flux divided by area: $B = {}^{\Phi}/_{A}$, and its unit is the tesla (Wb/m^2).





Force on a Conductor

- When a charge particle moves in a region with a magnetic field B, the particle will be deflected because of a magnetic force:

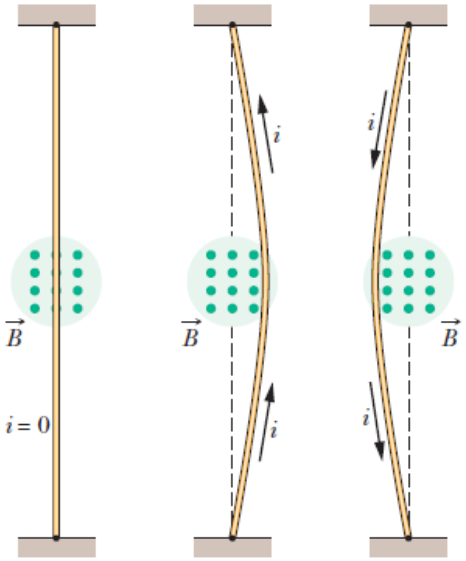
$$\overrightarrow{F_B} = q\vec{v} \times \vec{B}$$

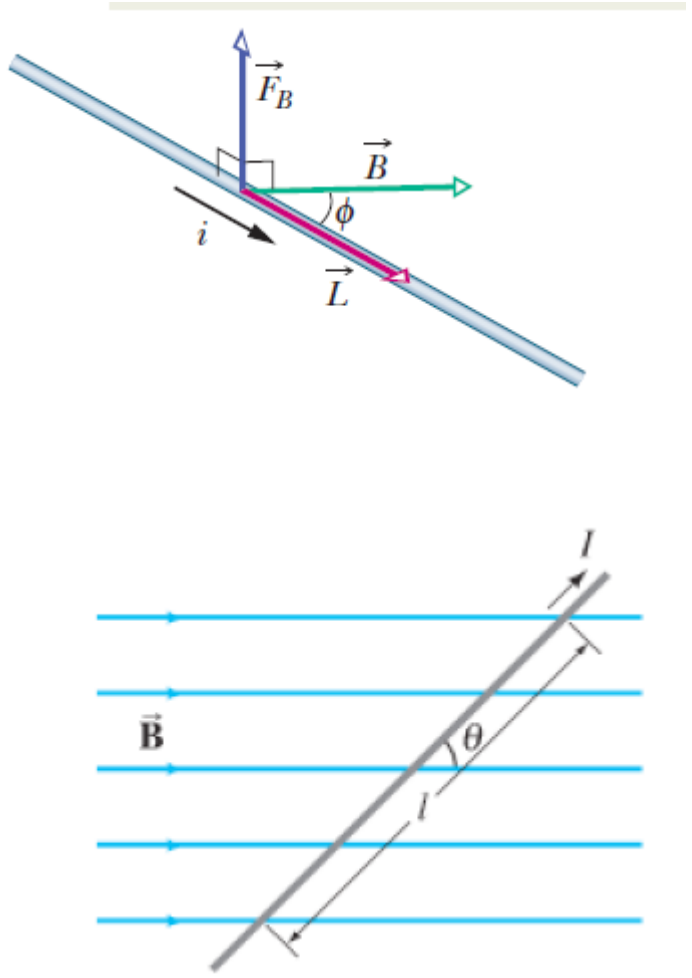
- Then, a conductor wire with a current in a magnetic field, will also be deflected by a magnetic force:

$$\overrightarrow{F_B} = i\vec{L} \times \vec{B}$$

- For practical purposes, we want the max magnetic force possible (when the wire is perpendicular to $\vec{B}$):

$$F_B = iLB$$



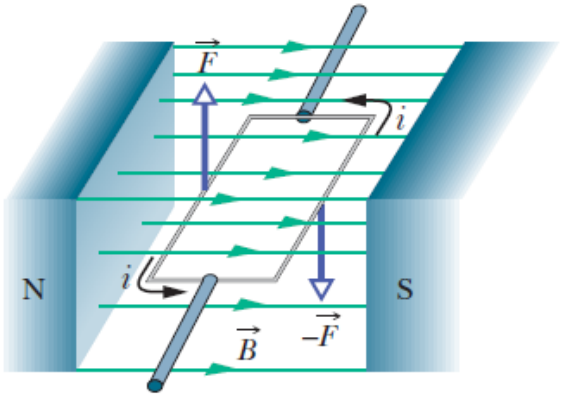


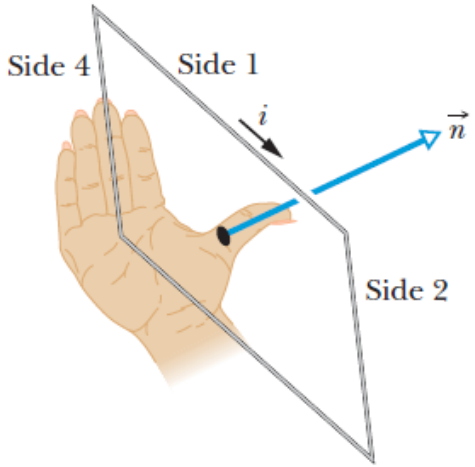
Torque on a Conductor Loop (Coil)

- A rectangular wire loop with a current immersed in a magnetic field will rotate.
- The torque that causes this rotation is given by:

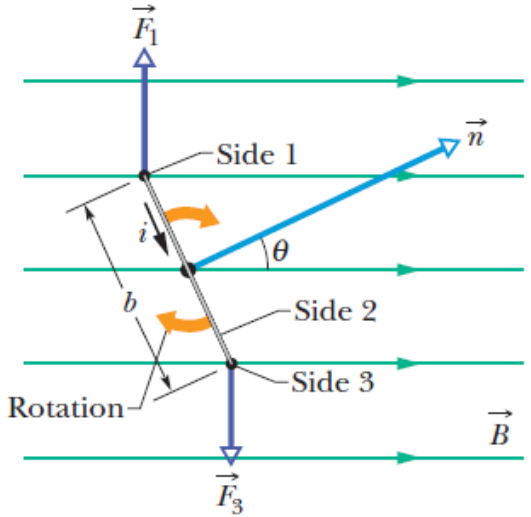
$$\tau = iabB \sin\theta$$

- Notice that $ab = A$, area of the loop
- Notice we can have N coils (loops) to increase the torque by N.





(b)



(c)

The Magnetic Dipole Moment

- If we define $\vec{\mu} = NiA\hat{n}$ (N is the number of turns and $\hat{n}$ is the unit vector parallel to $\vec{n}$), we can re-write the torque on the coils exerted by the magnetic field as:

$$\vec{\tau} = \vec{\mu} \times \vec{B}$$

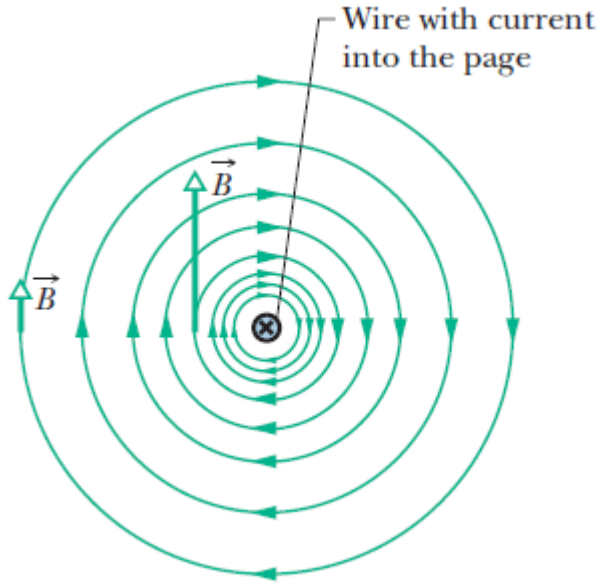
- $\vec{\mu}$ is called the magnetic dipole moment

Magnetic Fields Due to Currents

- Moving charged particles create a magnetic field around themselves. Current create magnetic fields.
- For example, a long wire with current i,
- The conductor loop,
- The solenoid.

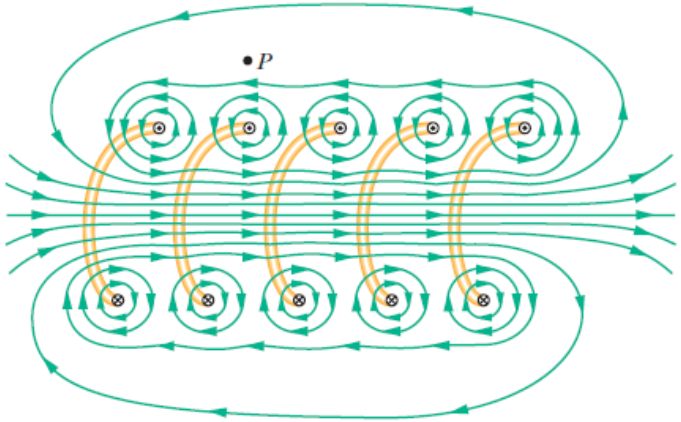
$$B = \frac{\mu_0 i}{2\pi R} \quad \text{(long straight wire).}$$

μ_0 is the vacuum ($\approx$ air) permeability



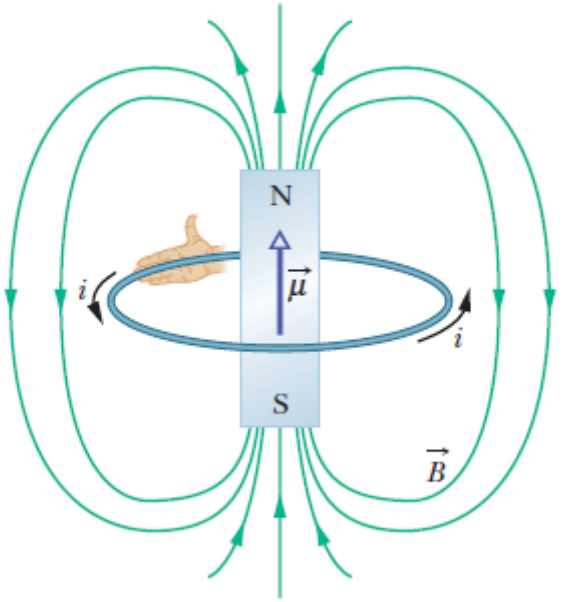
$$B = \mu_0 i n \quad \text{(ideal solenoid).}$$

(B inside the solenoid.
n = # of turns per unit length)



A Current-Carrying Coil as a Magnetic Dipole

- A coil (carrying a current) is a magnetic dipole similar to any magnet you are used to.
- The coil produces a magnetic field. We can apply the right-hand rule to find the direction of the magnetic field.
- The main idea here is that a coil with current behaves as a magnetized needle. When the coil is placed in an external magnetic field, it will rotate as a magnetized needle does.
- The rotation ceases when the external magnetic field and the magnetic field produced by the coil are aligned to each other.



Faraday's Law (Electromagnetic Induction)

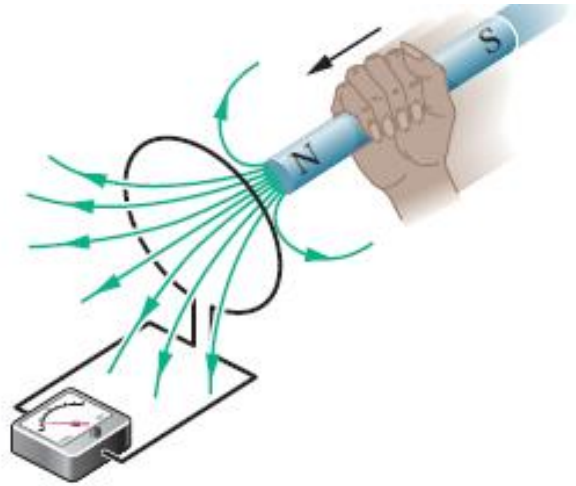
- A changing magnetic flux causes an electromotive force (voltage).
- If the magnetic flux that passes through a coil changes in time, then a voltage is induced in the terminals of the coil:

$$v = -\frac{d\Phi}{dt}$$

- For a coil with N turns:

$$v = -N\frac{d\Phi}{dt}$$

- The minus sign is often neglected. It means that the magnetic flux generated by the coil should oppose the rate of change of the external magnetic flux.
- Notice that when the coil rotates in our previous case, the flux through the coil changes, generating what we know as back electromotive force.



Magnetomotive Force

- The ability of a coil to produce flux is given by its magnetomotive force (mmf):

$$mmf = Ni$$

- mmf units are ampere-turns
- Reluctance $\mathcal{R}$ gives a measure of how difficult it is for the magnetic flux to complete its circuit. The “Magnetic Ohm’s law” is then:

$$\Phi = \frac{mmf}{\mathcal{R}} = \frac{Ni}{\mathcal{R}}$$

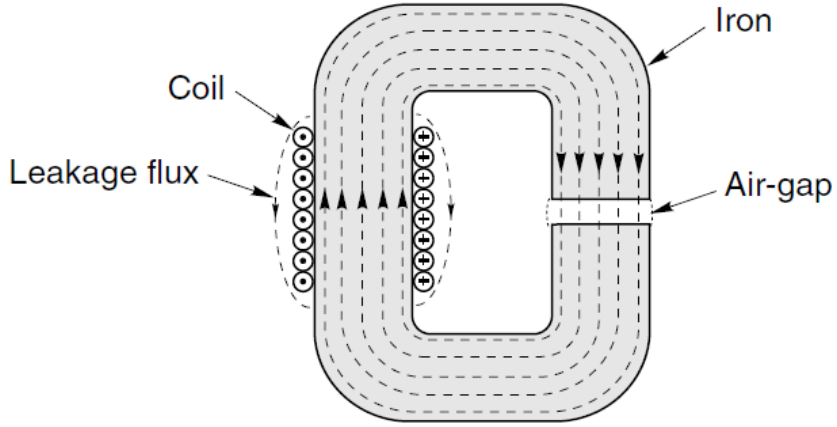
- A good (with high permeability) magnetic material has small reluctance. Air has a big reluctance, ferromagnetic material has small reluctance.

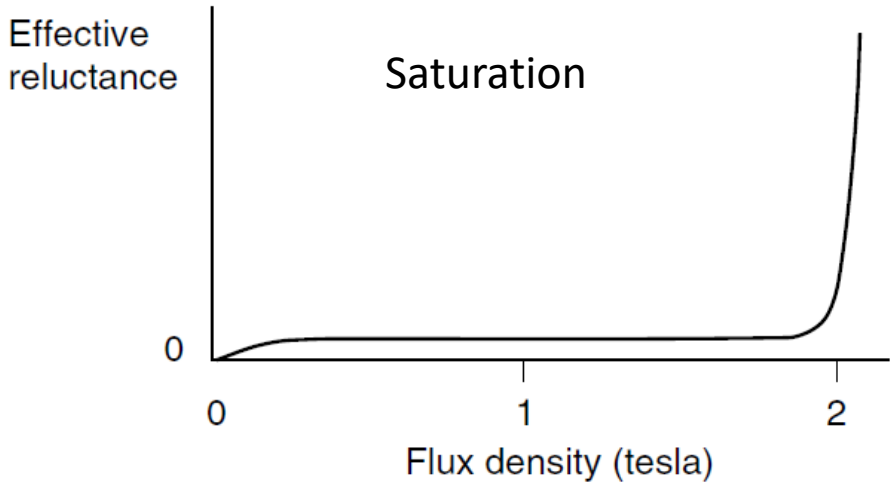
Magnetic Circuits

- In motors, irremediably, we will need air gaps. Air have high reluctance.
- The air gap has to be narrow to avoid flux losses:

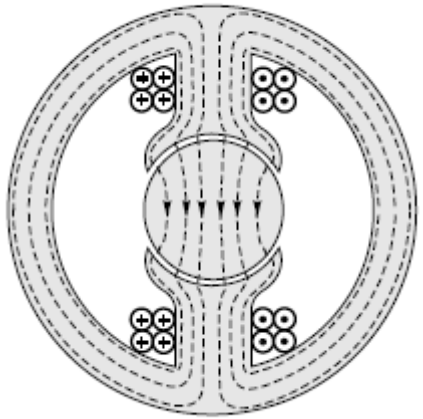
$$\mathcal{R}_g = \frac{d}{A\mu_0}$$

- d is the width of the gap and A is the area of the gap. The smaller the gap, the better.
- Saturation is the limit of flux that the any material has. At this point, increasing the mmf will not translate into increasing the flux. It's equivalent to say that the reluctance is too high at this point.

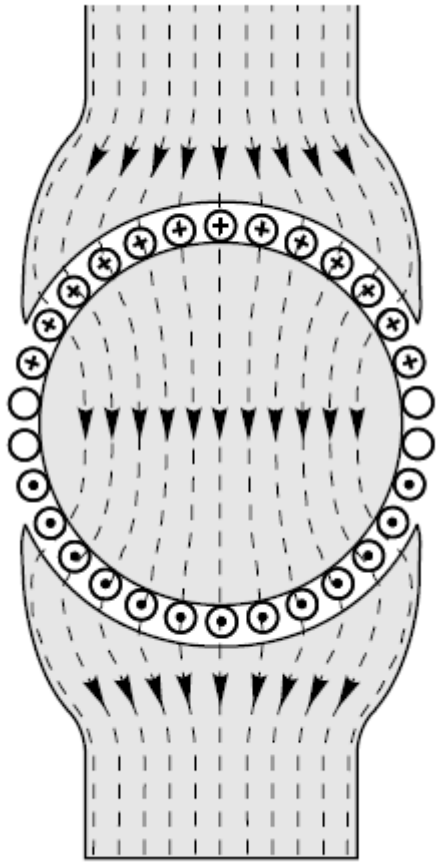




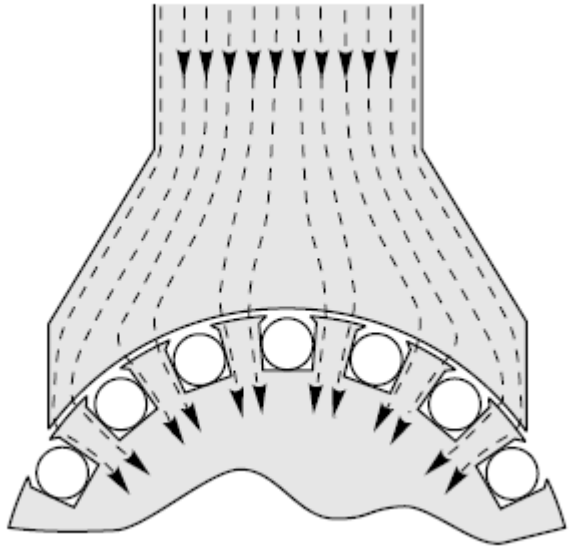
Magnetic Circuits in Motors



Typical Motor Magnetic Circuit

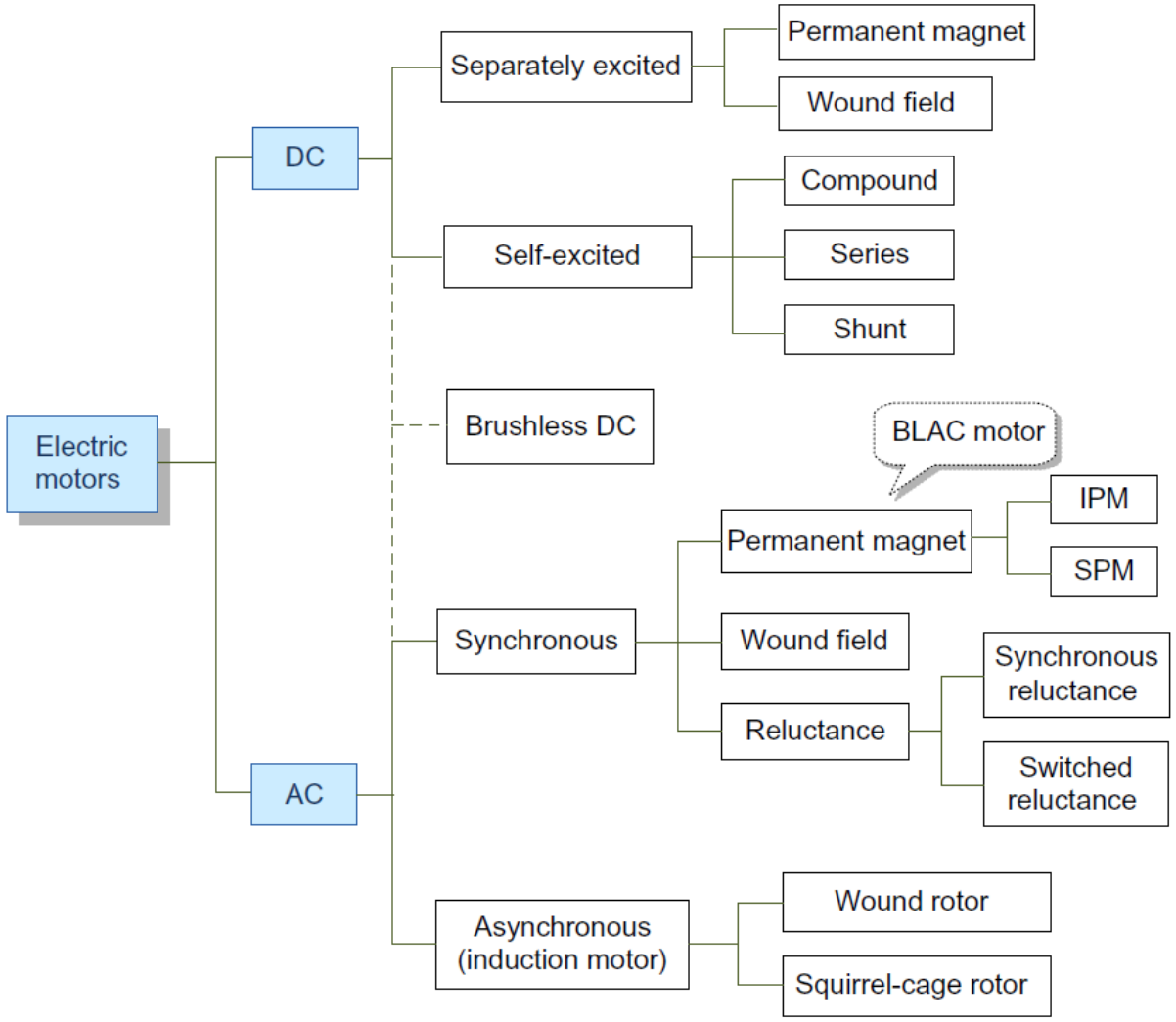


Wire arrangement in the rotor to produce torque



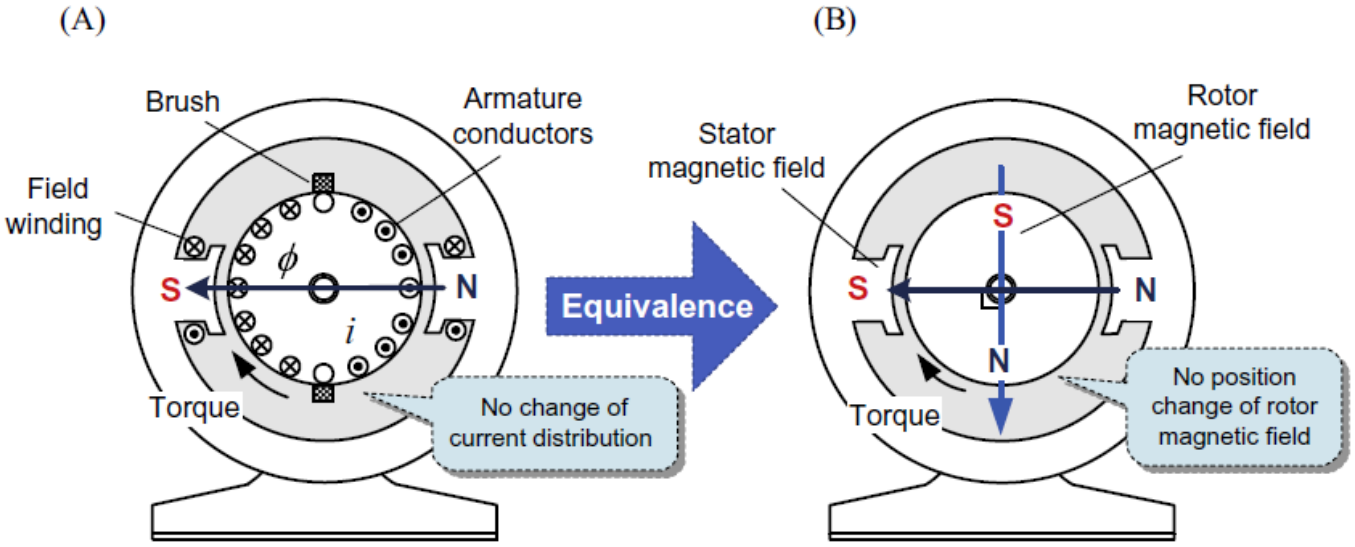
The rotor is slotted to accommodate conductors and guarantee rotor motion more efficiently.

Classification of Electric Motors

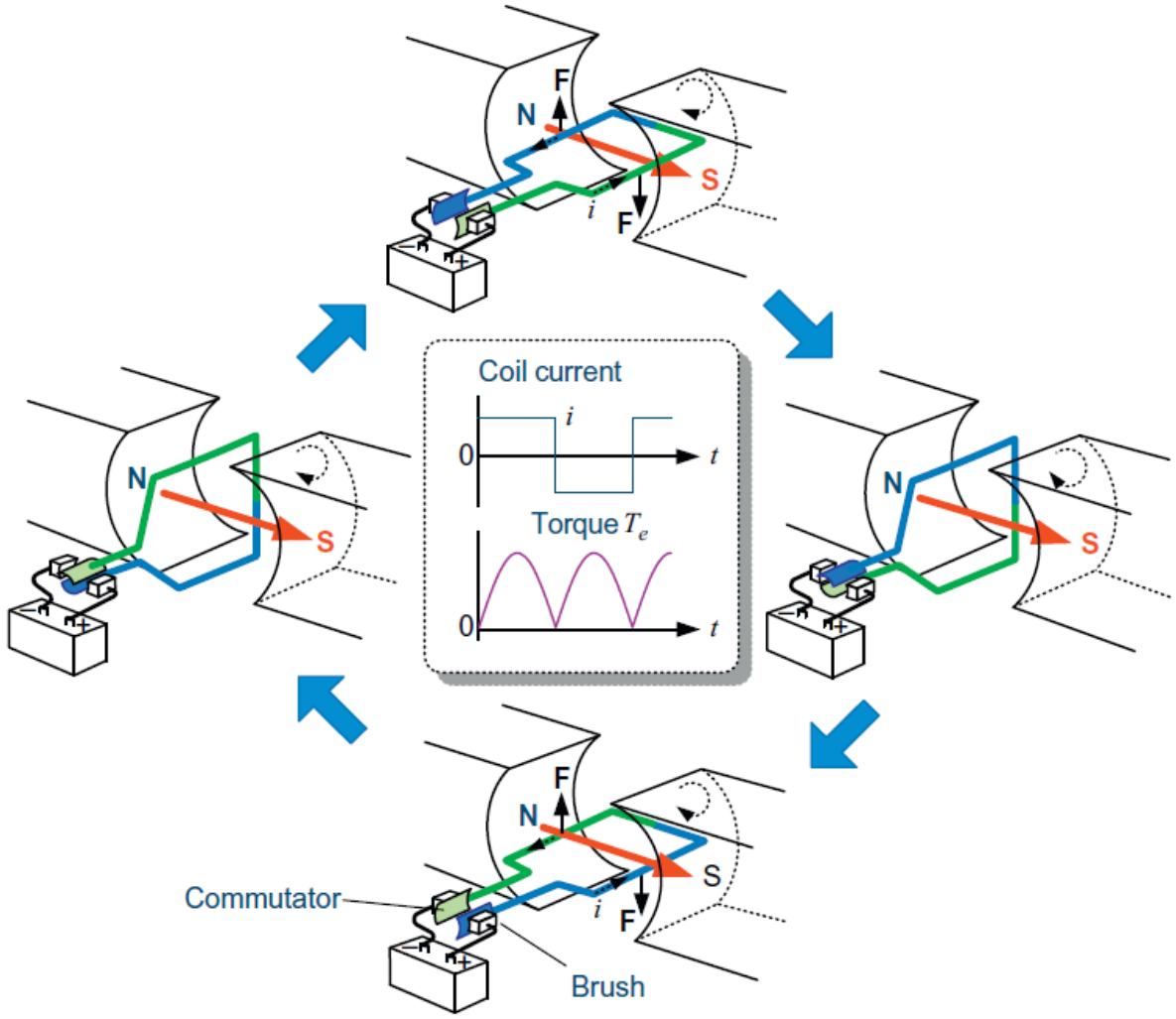


DC Motor Basic Principles

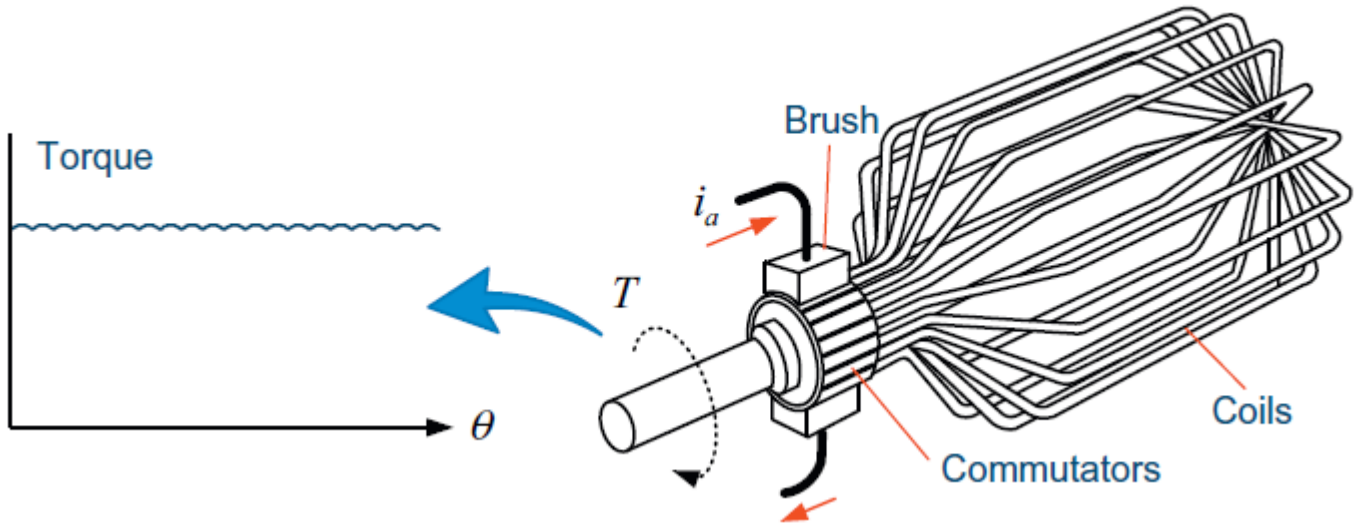
- There are two stationary magnetic fields in a DC motor.
- One Stationary magnetic field is the stator magnetic field produced by magnets or a field winding.
- The other is the rotor magnetic field produced by the current in the conductors of the rotor
- The rotor magnetic field is stationary despite the rotation of the rotor.
- This is achieved through brushes and commutators as explained later.
- Because these two magnetic fields are always perpendicular to each other, there's a continuous torque as predicted by the torque equation of previous slides



DC-Motors


F
N
S
i
F
Coil current
i
0
t
Torque T_e
0
t
N
S
N
S
F
N
S
i
F
Commutator
Brush

DC-Motors



DC-Motors – Lap Winding

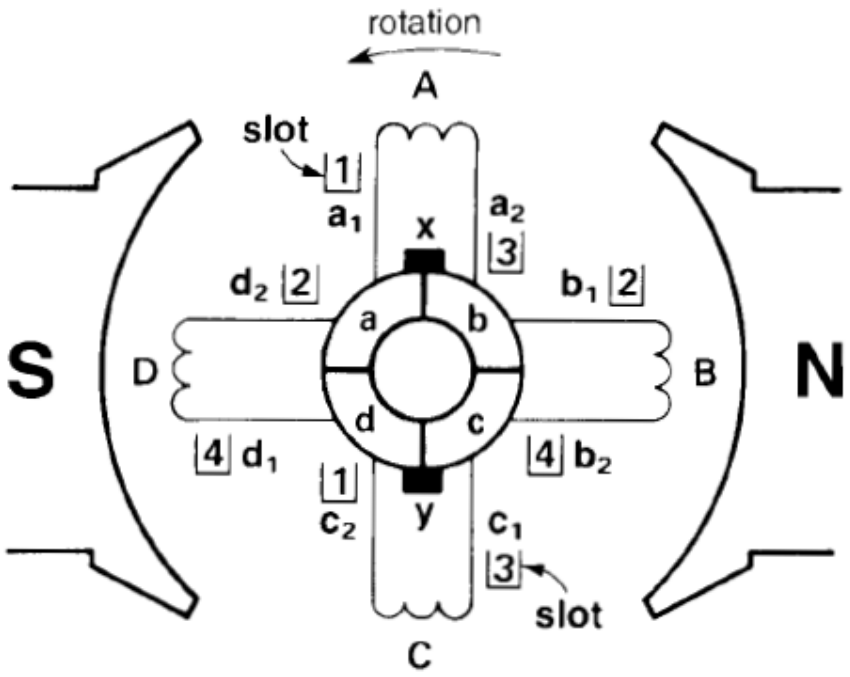


Figure 4.7
Schematic diagram of a dc generator having 4 coils and 4 commutator bars. See Fig. 4.9.

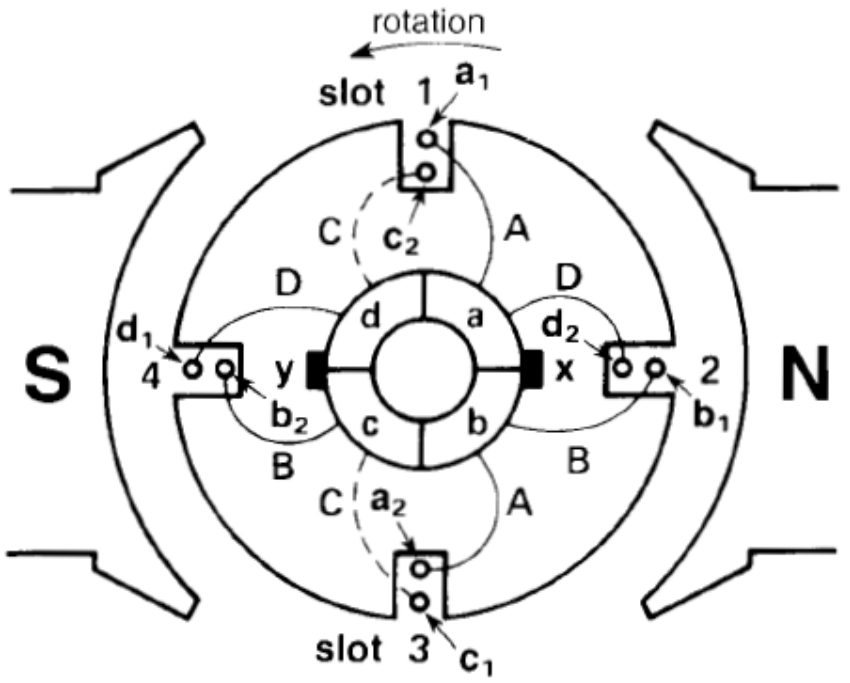
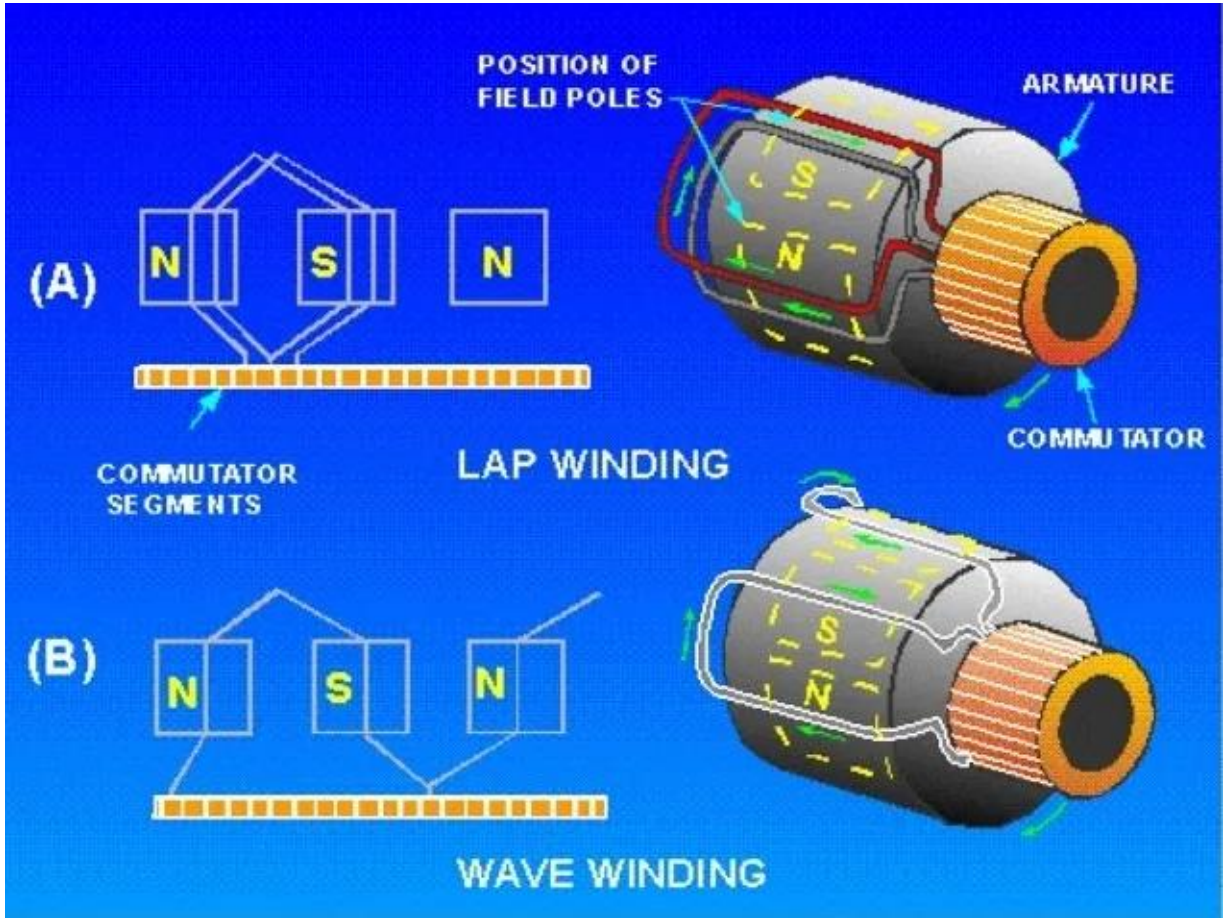
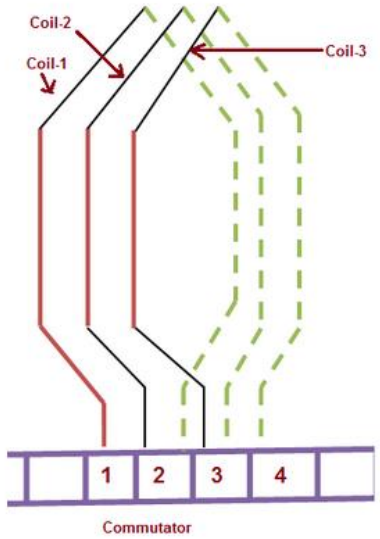
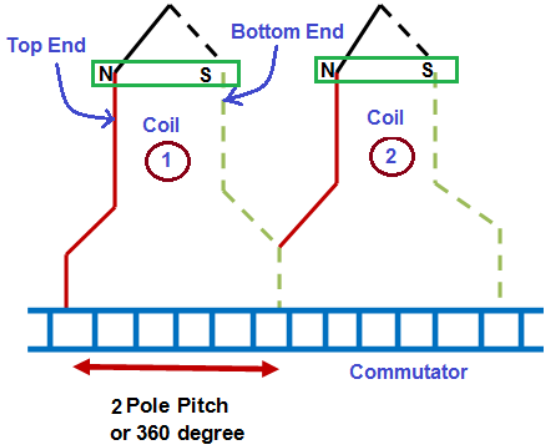


Figure 4.9
The actual physical construction of the generator shown in Fig. 4.7. The armature has 4 slots, 4 coils, and 4 commutator bars.

Lap and Wave Winding



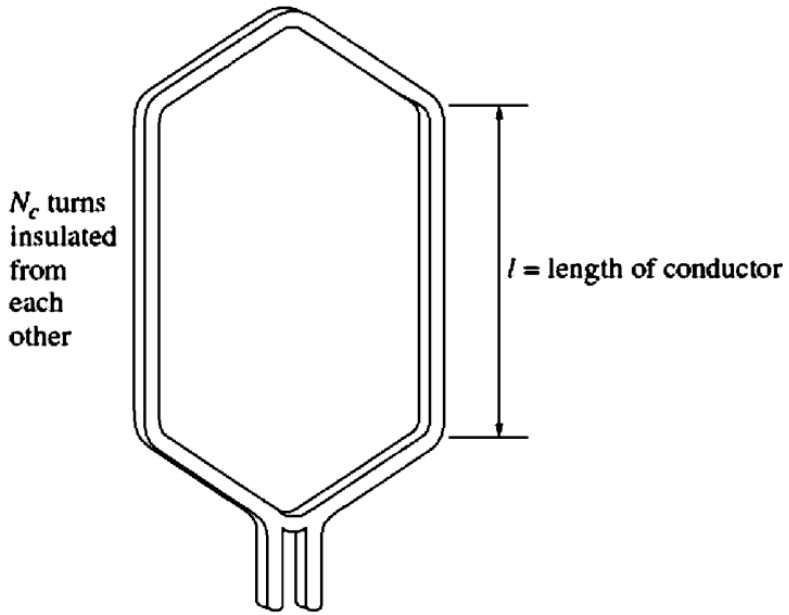




Lap and Wave Winding Differences

- Lap:
  - Number of parallel paths (a) = Number of poles (P)
  - Number of brushes = Number of poles
  - Used for low voltage – high current applications
  - The generated emf in lap winding is independent of number of poles
  - Usually More costly (more brushes, add equalizer rings)
- Wave:
  - Number of parallel paths = 2
  - Number brushes = 2, may be match the number of poles
  - Used for high voltage – low current applications
  - The generated emf depends on the number of poles

DC Rotor Coils



Regardless of the way in which the rotor windings are connected to the commutator segments, the rotor windings are diamond shaped coils which are inserted into the armature slots as a unit. Each coil consists of Nc turns (loops), each turn insulated from the others and the rotor slot, . Each side of a coil is called a *conductor*. Then, the number of conductors, Z, on a machine armature is given by:

$$Z = 2CN_C$$

Z = number of conductors on rotor
C = number of coils on rotor
Nc = number of turns per coil

DC Machines

GENERATED VOLTAGE OF DC MACHINES

The Voltage depends On:

- The flux ϕ in the machine
- The angular speed w of the machine's rotor
- A constant depending of the construction of the machine

$$E_b = K\phi w$$

$$K = \frac{ZP}{2\pi a}$$

$$\phi = B\left(\frac{2\pi r l}{P}\right)$$

Where:

$$E_b = Induced\ Voltage\ (back\ emf)$$
$$\phi = flux\ per\ pole$$
$$w = rotor's\ angular\ velocity$$
$$K = constant\ that\ depends\ on\ the\ motor$$
$$Z = number\ of\ conductors$$
$$P = number\ of\ poles$$
$$a = number\ of\ parallel\ (current)\ paths$$
$$(number\ of\ conductors\ per\ slot)$$
$$l = length\ of\ conductor$$
$$r = average\ radius\ of\ coil$$
$$B = magnetic\ field$$

DC Machines

GENERATED TORQUE OF DC MACHINES

The torque depends on:

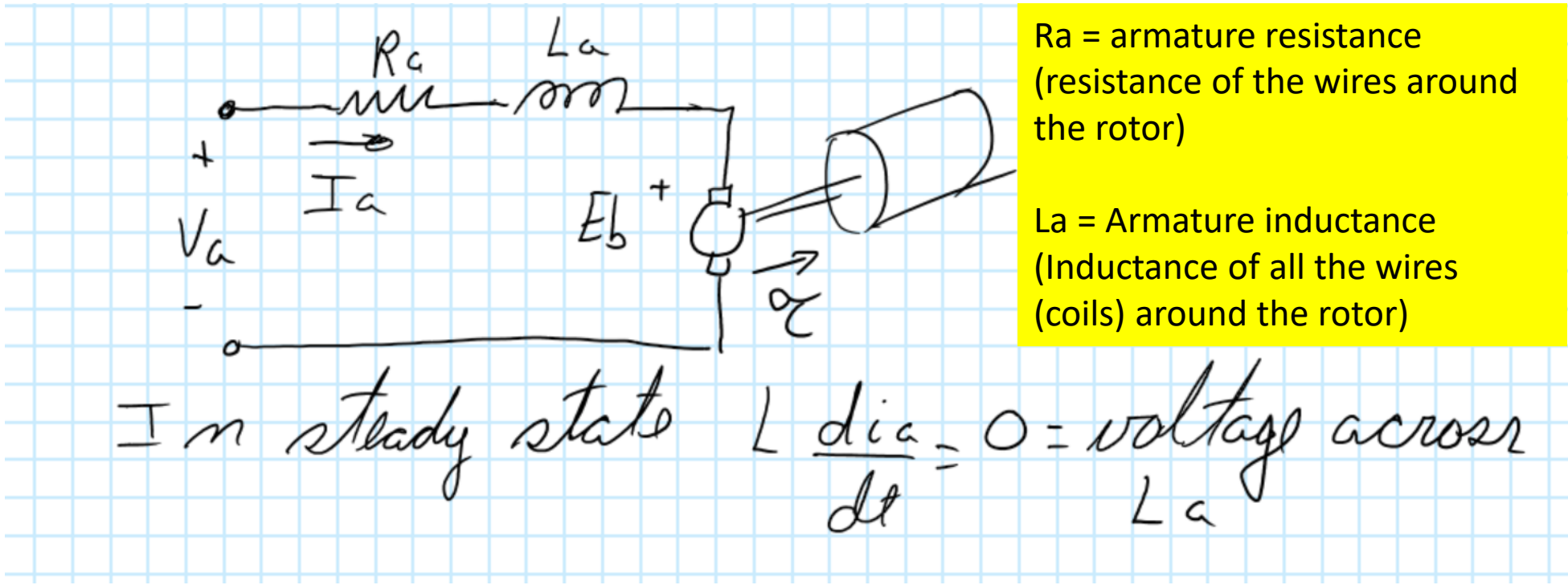
- The flux in the machine
- The armature (or rotor) current in the machine
- A constant depending on the construction of the machine

$$\tau = K\phi I_A$$

$$K = \frac{ZP}{2\pi a}$$

$$\phi = \frac{2\pi r l}{P} B$$

DC Machine Schematic



Ra = armature resistance (resistance of the wires around the rotor)

La = Armature inductance (Inductance of all the wires (coils) around the rotor)

DC machines (Problems)

1. Calculate the voltage induced in the armature winding of a 4-pole, lap-wound, DC machine having 728 active conductors and running at 1800 rpm. The flux per pole is 30mWb.
2. What is the voltage induced in the machine of problem (1) if the armature is wave-wound?
3. Calculate the electromagnetic torque developed by the machine of problem (1) if the resistance of the conductors is 10 ohm and the input voltage is 800 V.
4. The wave-wound armature of a 6-pole, 600 rpm, has 90 coils. Each coil has 4 turns and the flux per pole is 0.04 Wb. Calculate the value of the induced voltage.

DC machines (Problems)

The following details are given on a 225 kW (≈ 300 hp), 250 V, 1200 r/min dc motor (see Figs. 5.4 and 5.5):

armature coils	243
turns per coil	1
type of winding	lap
armature slots	81
commutator segments	243
field poles	6
diameter of armature	559 mm
axial length of armature	235 mm

Calculate

a. The rated armature current (Assume $R_a, B_m \approx 0$)

b. The number of conductors per slot

c. The flux per pole

d. The flux density (B)

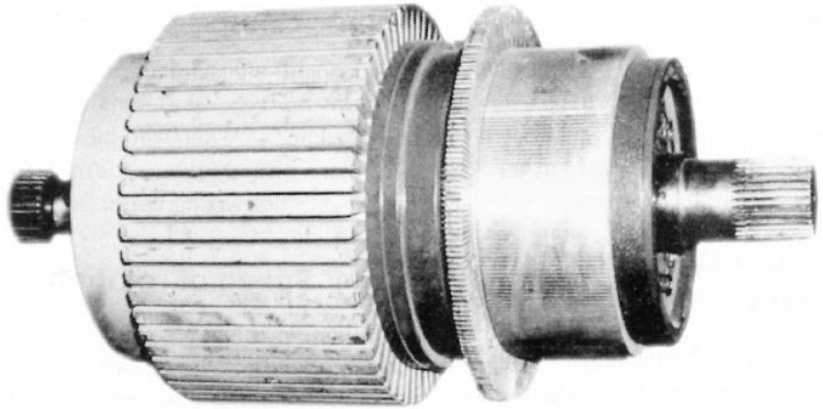


Figure 5.4
Bare armature and commutator of a dc motor rated 225 kW, 250 V, 1200 r/min. The armature core has a diameter of 559 mm and an axial length of 235 mm. It is composed of 400 stacked laminations 0.56 mm thick. The armature has 81 slots and the commutator has 243 bars. (*H. Roberge*)

DC Machines (Problems)

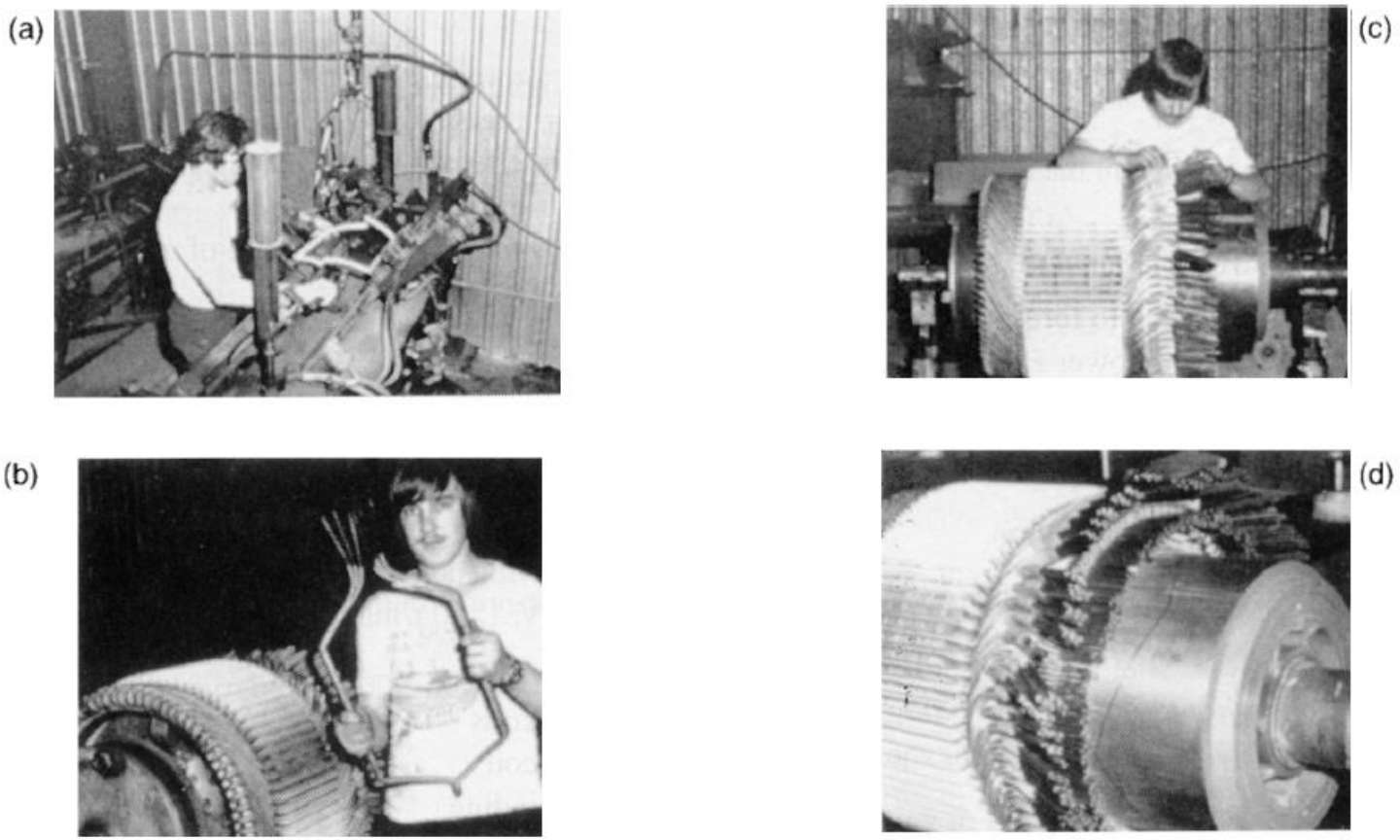


Figure 5.5
a. Armature of Fig. 5.4 in the process of being wound; coil-forming machine gives the coils the desired shape.
b. One of the 81 coils ready to be placed in the slots.
c. Connecting the coil ends to the commutator bars.
d. Commutator connections ready for brazing. (H. Roberge)

Solution:

- Because Ra is approx. 0, the induced voltage E_b (back emf) is nearly equal to the applied voltage (rated voltage) and the applied power at the input is available to the mechanical load (since Bm =0):

$$P_{out} = 225000\ \text{W} = I_a E_b = I_a(250\ \text{V}) \Rightarrow \boxed{I_a = 900\ \text{A}}$$

- Each coil has 1 turn and 2 conductors, so altogether there are

$$\#\ \text{Conductors} = \text{Z} = 243 \times 1 \times 2 = 486$$

$$\boxed{\text{Conductors per slot} = 486/81 = 6 = \text{a}}$$

- Recall that the conductors per slot is the number of parallel paths (a) that should match the number conductors per slot

Solution (Cont…)

- To find the flux per pole we need to find km:

$$k_m = K\Phi \qquad (1)$$

- We know that:

$$E_b = k_m \omega$$

$$250\ \text{V} = k_m \left(\frac{1200 \times 2\pi}{60\ \text{s}} \right)$$

$$k_m = \frac{6.25}{\pi}\ \text{V}-\text{s} \qquad (2)$$

Solution (Cont…)

- We find now K (lap winding):

$$K = \frac{ZP}{2\pi a} = \frac{486 \times 6}{2\pi \times 6} = \frac{243}{\pi} \qquad (3)$$

- Using (1), (2) and (3):

$$\Phi = \frac{k_m}{K} = \frac{6.25 \times \pi}{\pi \times 243} = 0.0257 \text{ Wb}$$

- To find B:

$$\Phi = B\frac{2\pi r l}{P} = B\frac{2\pi(\frac{0.559}{2} \text{ m})(0.235 \text{ m})}{6} \Rightarrow \boxed{B = 0.374 \text{ T}}$$

Problem

- The rated values for a separately excited DC motor are 5KW, 140 V, 1800 rpm, and its efficiency (η) is 91%. The armature resistance is 0.25 ohm. Find (a) The rated armature current and torque, (b) the speed at full load when the armature voltage is reduced by 10%, (c) the speed at full load current when the flux is reduced by 10%, and (d) the output torque available by armature current at this speed.

Solution

- a) Because of the efficiency, we have

$$\eta = \frac{P_{out}}{P_{in}} \Rightarrow P_{in} = \frac{5000}{0.91}\ \text{W} = 5494.51\ \text{W}$$

$$P_{in} = V_R I_R \Rightarrow \boxed{I_R = \frac{5494.51\ \text{W}}{140\ \text{V}} = 39.25\ \text{A}}$$

- a) The power at the load is

$$P_{out} = T_R \omega_R \Rightarrow \boxed{T_R = \frac{5000\ \text{W}}{(1800 \times 2\pi)/60\ \text{s}} = 26.53\ \text{N}-\text{m}}$$

Solution (Cont…)

- b) At full load $I_a = I_R$, we have

$$\omega = \frac{E_b}{k_m} = \frac{0.9V_R - R_a I_R}{k_m} \quad (1)$$

- b) We can find k_m by knowing that

$$k_m = \frac{E_{bR}}{\omega_R} = \frac{V_R - R_a I_R}{\omega_R} = \frac{140\ \text{V} - (0.25\Omega)(39.25\text{A})}{(1800 \times 2\pi)/60\ \text{s}} = 0.691\ \text{V–s}$$

- b) Replacing k_m in (1)

$$\omega = \frac{116.188\ \text{V}}{0.691\ \text{V–s}} = 168.2253\ \text{s}^{-1} = 1606\ \text{rpm}$$

Solution (Cont…)

- c) Since $k_m = K\phi$, a 10% flux reduction is equivalent to a 10% of k_m reduction: $k'_m = 0.9k_m$

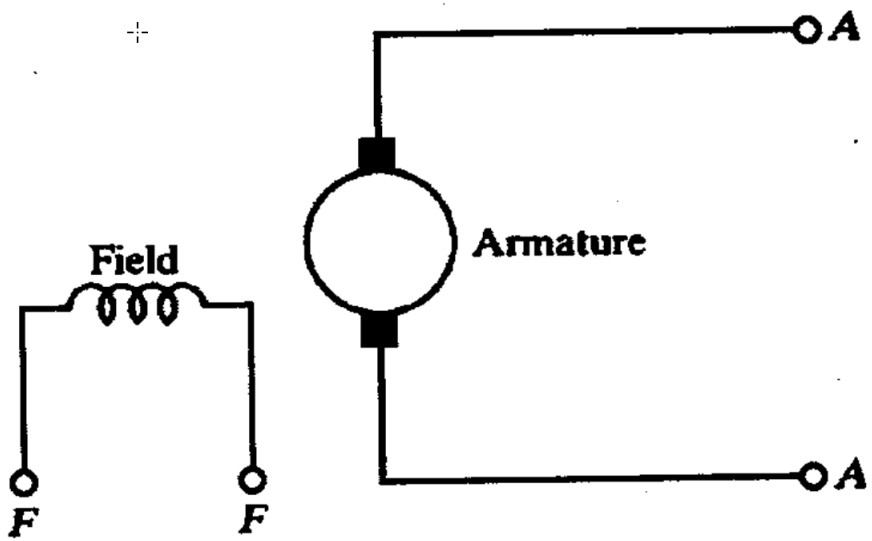
$$\omega = \frac{V_R - R_a I_R}{0.9k_m} = \frac{140\ \text{V} - (0.25\Omega)(39.25\text{A})}{0.9 \times 0.691} = 209.34\ \text{s}^{-1}$$

- c) $\boxed{\omega = 209.34\ \text{s}^{-1} = 1999\ \text{rpm}}$

- d) The new available torque from the armature is

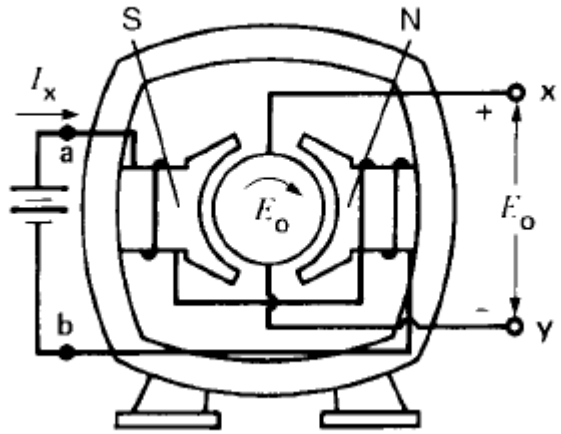
$$\boxed{T' = k'_m I_R = 24.41\ \text{N}-\text{m}.}$$

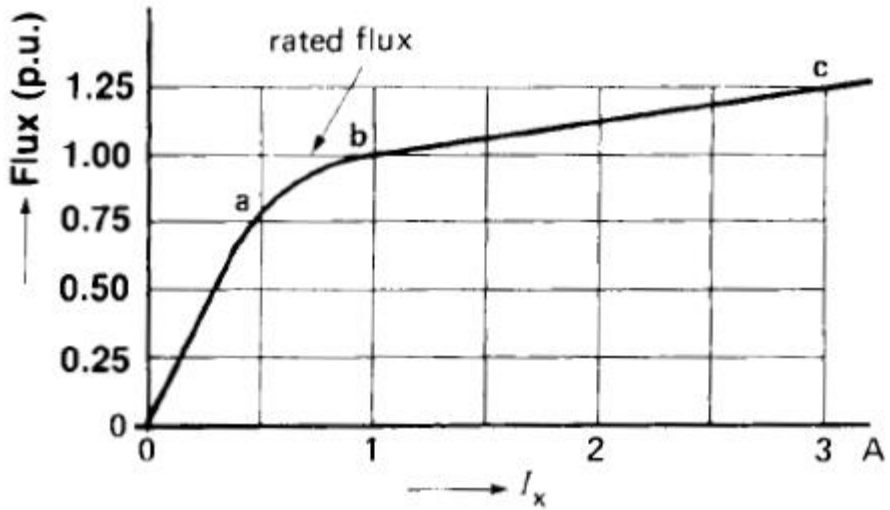
DC Machines (Separately Excited)



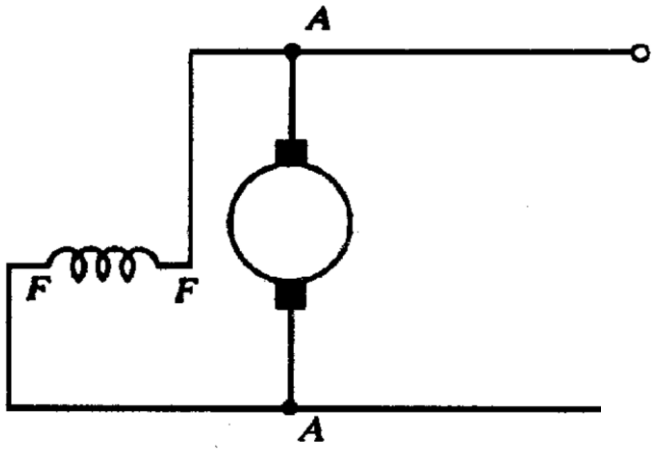
(a) Separately excited

Fig. 4-8(a)

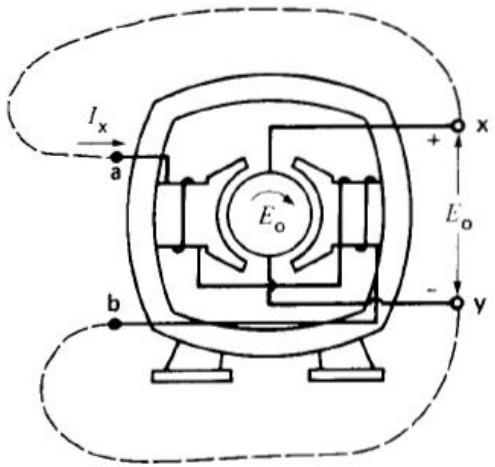


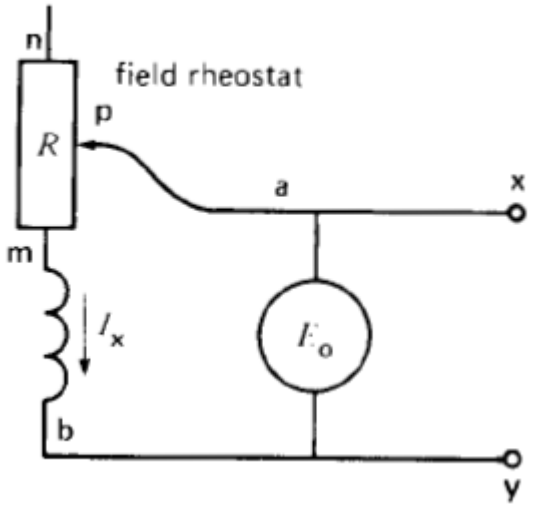


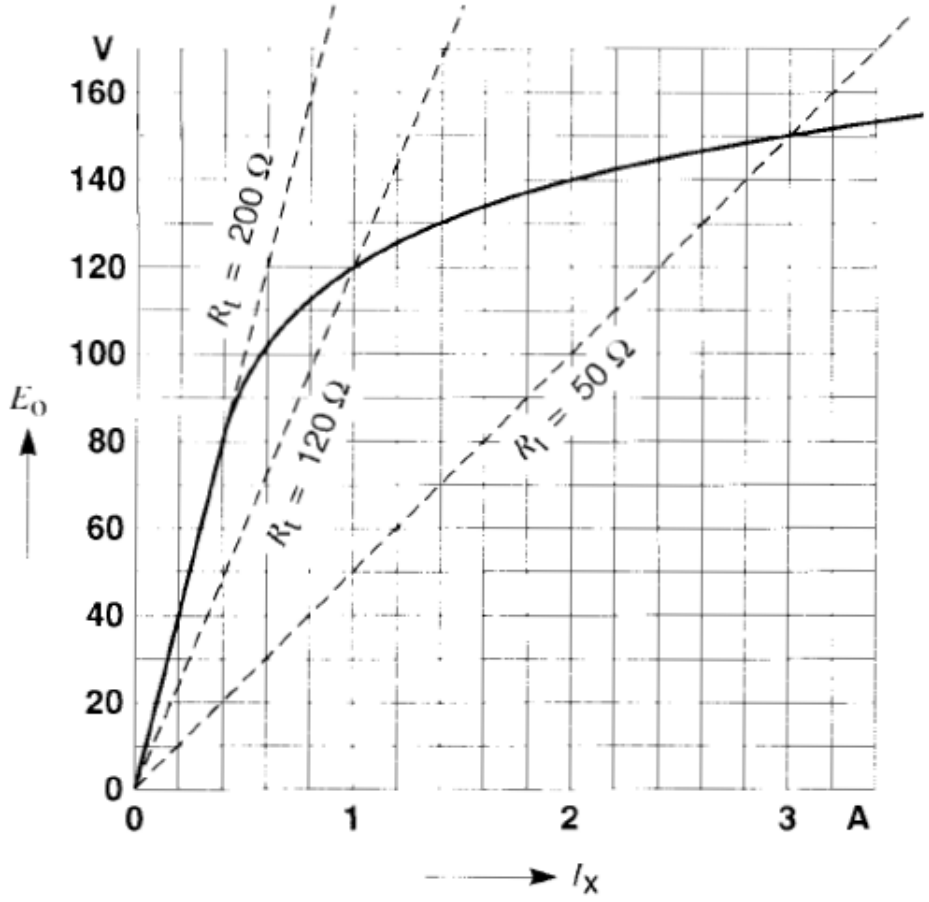
DC Machines (Shunt)



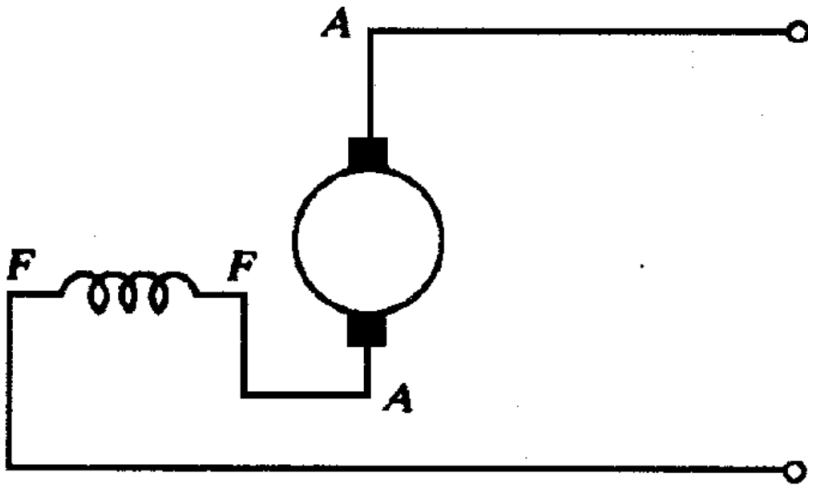
(b) Shunt





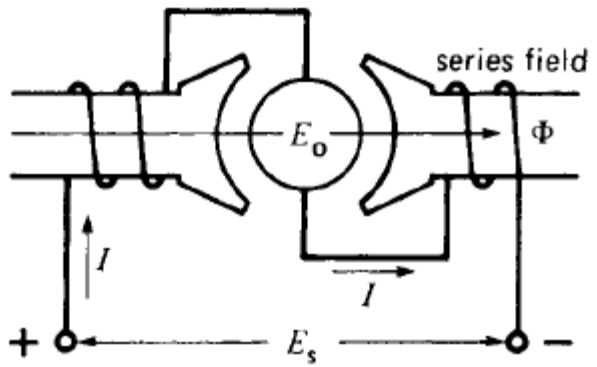


DC Machines (Series)



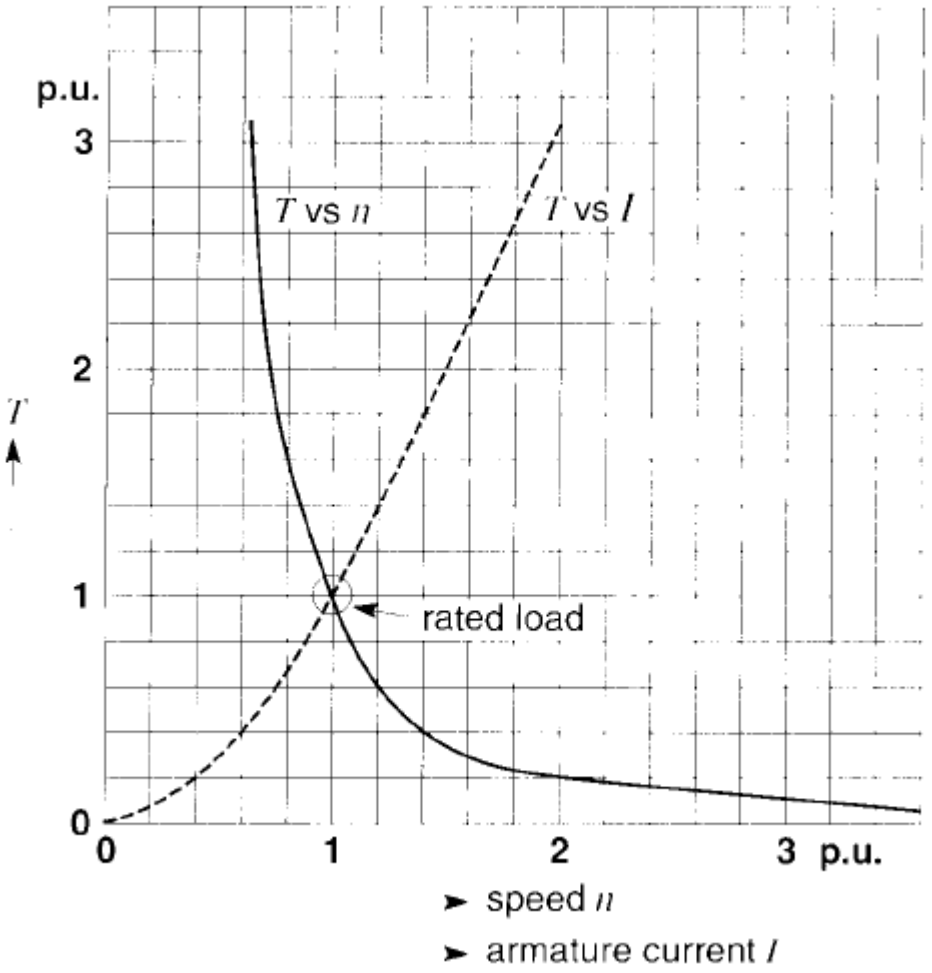
(c) Series

(a)



DC Machines (Problem)

- A 15 hp, 240 V, 1780 rpm dc series motor has a full load current of 54 A. Its operating characteristics are given by per-unit curves in the figure. (a) The current and speed when the load torque is 24 N-m. (b) The efficiency in these conditions and with V_R.



Solution

- A per-unit characteristic is a curve (or equation) normalized by a base, in this case the rated values. Therefore, a speed of 1 in per-unit, means the speed is the rated value. A current of 0.9 means the current is 0.9 times the rated value and so on.
- We have $P_R = 15\ hp = 15 \times 746 = 11190\ W$; $\omega_R = 1780\ rpm = 186.40\ rad/s$; $T_R = P_R/\omega_R = 60.03\ N-m$.
- a) A load torque of 24 N-m corresponds to T(pu) =24/60.03 = 0.4. Using the curve T(pu) vs. $\omega(pu)$, we can find $\omega(pu)$: $\omega(pu) = 1.4 \Rightarrow$ $\boxed{\omega = (1.4) \times (1780\ rpm) = 2492\ rpm = 261\ rad/s}$.

Solution (Cont…)

- a) Using the curve T(pu) vs I(pu), a per unit torque of 0.4 corresponds to a per-unit current of 0.6. That is,

$$I = 0.6 \times I_R = 0.6 \times 54 = 32.4\ A$$

- b) To calculate the efficiency, we have

$$\eta = \frac{P_{out}}{P_{in}}$$

$$P_{in} = V_R I = (240)(32.4) = 7776\ W$$

$$P_{out} = \omega T = (261)(24) = 6264\ W$$

$$\eta = \frac{6264}{7776} = 0.8055\ (80.55\%)$$